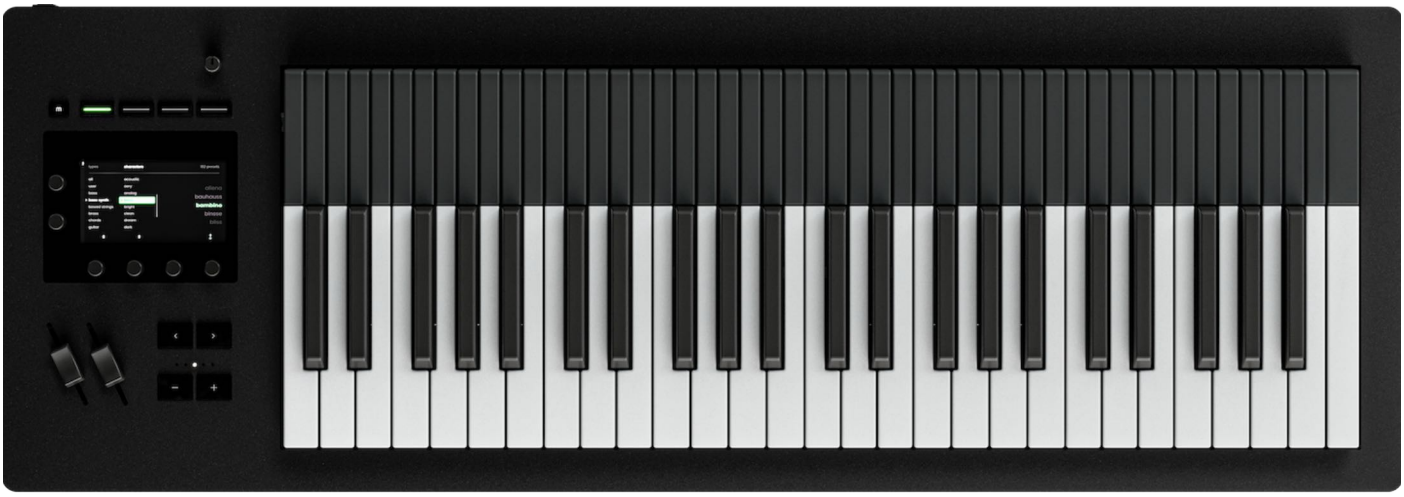
BITWIG STUDIO CONTROLLER EXTENSION

Expressive E
Osmose 49/61

BITWIG

Setup

This document gives you a brief overview of how to set up and use Osmose with Bitwig Studio. For an in-depth explanation of all Osmose features, please read the Osmose user guide provided by Expressive E.



Important: Please make sure that you use Osmose firmware version 2.3.14 or later.

Osmose should be detected and set up automatically when it is connected. If you added Osmose manually, make sure the MIDI ports are selected in the correct order.

macOS:



Windows:



DAW Control Mode

1 Enter DAW Control Mode
When Bitwig is started, the DAW control menu is shown. Select *daw control* and press an encoder to confirm.

2 Open Main Menu
Hold SHIFT and press Encoder 1 to return to the Main Menu.

3 Select Preset/Undo
Press < to select the previous preset of the Osmose CE.

Hold SHIFT and press < to undo the last operation.

4 Select Preset/Redo
Press > to select the next preset of the Osmose CE.

Hold SHIFT and press > to redo the last operation.

5 Octave Down/Quantize
Press - to move the octave range downward.

Hold SHIFT and press - to quantize the currently selected clip.

6 Octave Up/Set Tempo
Press + to move the octave range upward.

Hold SHIFT and press + to open the tempo page. Set the BPM of the project either by turning one of the encoders below the screen or by tapping the + button repeatedly at the desired tempo. Press the *m* button on the upper left to close the tempo page.

7 Pitch
Move the fader to send pitch bend messages.

8 Modulation
Move the fader to send modulation messages (CC1).



Transport Functions

1 Open Transport Page
Press *m* to show or hide the transport page.

2 Play/Pause
Press PLAY to start the transport, press PLAY while the transport is active to pause playback.

3 Stop/Return to Zero
Press STOP to stop the transport, press it a second time to return the playhead to position 1.1.1.0

4 Record
Press RECORD to activate recording.

5 Metronome
Press METRONOME to toggle the metronome on or off.

Encoder A controls various parameters:
6 Playhead Position
Turn the encoder to move the playhead in the Arranger.

6 Loop Position/Length
Push the encoder to toggle the encoder between playhead and loop control.

When loop is active, turn the encoder to move the loop position in the Arranger.

Hold SHIFT and turn the encoder to adjust the length of the loop.

6 Add/Jump to Marker
Hold SHIFT and press the encoder to add a marker at the current playhead position.

Hold SHIFT and turn the encoder to jump to the previous or next marker in the Arranger.

Encoder B controls various parameters:
7 Select Track
Turn the encoder to select the previous or next track.

7 Open Plug-in Window
Press the encoder to open or close the plug-in window (VST) or the expanded device view of a Bitwig Studio Device (if available).

7 Select Scene
Hold SHIFT and turn the encoder to select the previous or next clip (or empty slot) of a scene.

7 Launch Clip
Hold SHIFT and press the encoder to launch the currently selected clip. If the selected slot is empty and the track is record enabled, a new clip will be recorded.

8 Track Volume/Mute
Turn the encoder to adjust the volume of the currently selected track.

Press the encoder to toggle its mute status.

9 Pan Position/Solo
Turn the encoder to adjust the pan position of the currently selected track.

Press the encoder to toggle its solo status.

10 FX Send 1
Turn the encoder to adjust the FX Send 1 level of the currently selected track.

11 FX Send 2
Turn the encoder to adjust the FX Send 2 level of the currently selected track.



Remote Controls

1 Open Adjust Page
Press button 2 to open the *adjust* page. When the *macros* tab is selected, the encoders will control the Remote Controls in Bitwig Studio.

2 Select Tab
To control the Remote Controls of the currently selected device in Bitwig Studio, make sure the *macros* tab is selected.

3 Remote Controls 1-4 (5-8)
Turn the encoders to control the Remote Controls of the currently selected device in Bitwig Studio.

By default, the four encoders control Remote Controls 1-4. Press the encoders to toggle them between controlling Remote Controls 1-4 and 5-8.

4 Select Device
Hold SHIFT and turn encoder 4 to select a device in the device chain of the currently selected track.

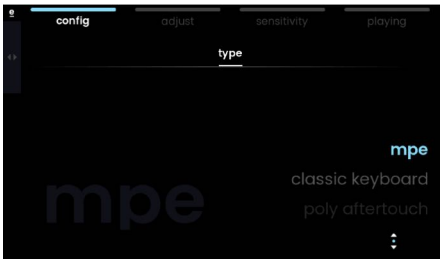


Osmose as an Input Device

Osmose can be used as an input device to play standard or MPE-enabled software instruments. Osmose's MIDI Port 1 (*Osmose 49/61 play* on macOS, *Osmose 49/61* on Windows) must be used for this task exclusively.

Choose a Configuration Preset for Osmose's External MIDI Mode

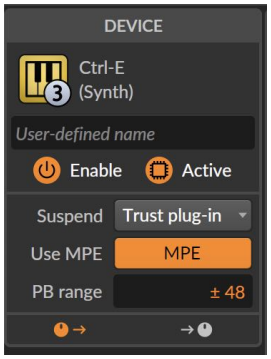
On Osmose, enter the config menu of External MIDI mode and select the configuration preset you'd like to use, for example: *classic keyboard* for standard plug-ins or *mpe* for MPE-enabled instruments.



Play Software Instruments with Osmose

Select an empty instrument track or create a new one. Add either one of Bitwig's internal instrument devices or a VST plug-in to the track. Click on the input selector of the track in the inspector panel or the mixer and select the first MIDI port of Osmose (*Osmose 49/61 play)* as the input. If the default setting *All ins* is used, controller data will be recorded twice, because Osmose sends data on both of its ports simultaneously.





For MPE-compatible VST plug-ins you have to activate MPE mode: Click on the MPE plug-in to show the device inspector, then click *MPE* next to *Use MPE* to activate MPE for the plug-in. For Bitwig devices this option is not available, it works automatically.

Configure MPE Interpretation in the Plug-in

Be aware that for some MPE-compatible plug-ins, MPE functionality must also be activated from within the plug-in. Check your plug-in's manual to learn about its MPE workflow.

When MPE is activated in the plug-in, *Pressure* will be controlled by Osmose Initial Pressure axis and *CC74 (MPE Timbre/Slide)* is controlled by Osmose's Aftertouch axis. For more details, see config menu.

In most MPE synths, the maximum MPE note pitch bend range (Axe X) is set to +/-48 semitones by default. Adjust the *bending range* fraction on Osmose (Ext MIDI mode > sensitivity menu > bending tab) for a more playable range. For example: When the maximum pitch bend range in your external synth is set to 48 semitones, set the fraction in Osmose to *1/48* to end up with a +/-1 semitone range for note bend gestures.

MPE Modulators in Bitwig Studio

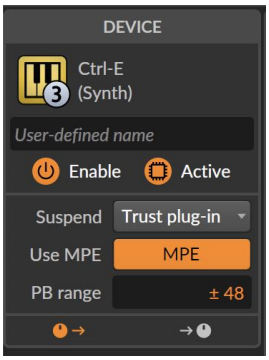
You can easily enhance presets of devices or VST plug-ins with MPE messages from Osmose by using Bitwig Studio's modulation system. Add an **Expressions** or a **MIDI** modulator to a modulator slot and map MPE messages to one or multiple parameters. The modulation mappings will be shown in the inspector panel. Read more about Bitwig Studio's modulation system in the *Bitwig Studio User Guide, chapter 15.2.*

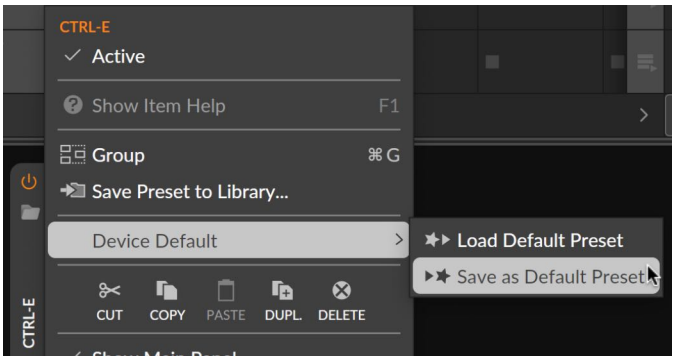
Using Ctrl-E

To use the full potential of the Ctrl-E plug-in in Bitwig Studio, you should activate the MPE setting of the plug-in, create a Remote Controls page and save this as the new default setting, so you only have to do this once.

Activate MPE Mode

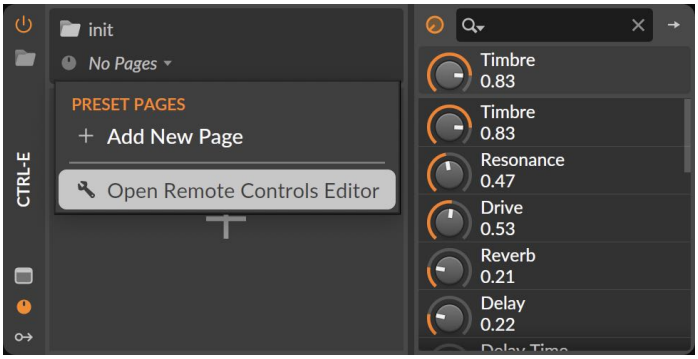
- Insert Ctrl-E on an instrument track in Bitwig Studio
- Click on Device header of Ctrl-E to select it
- In the Inspector on the left side, activate the *Use MPE* setting
- Right-click on the Device header of Ctrl-E and open the *Device Default* submenu
- Click on *Save as Default Preset*

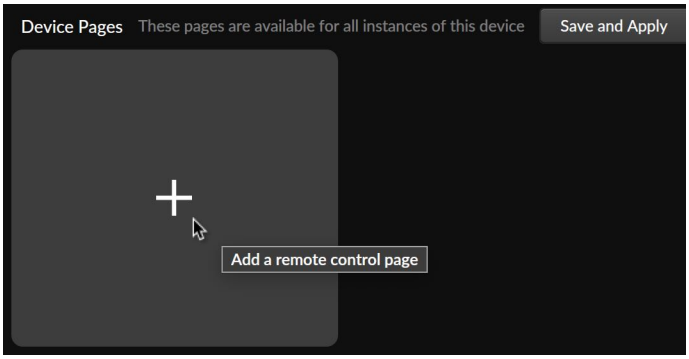




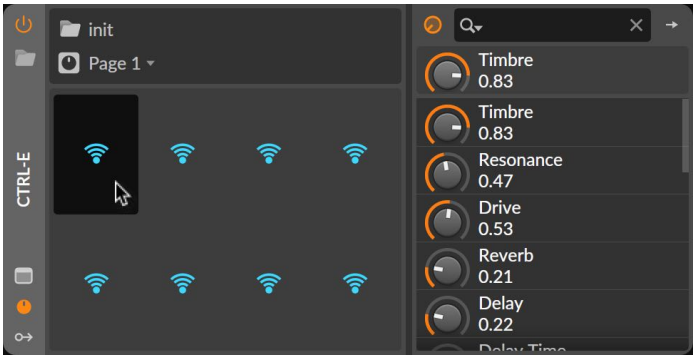
Create a Global Device Remote Controls Page

- Click on the Remote Controls icon in the Device header of Ctrl-E to show the Remote Controls
- Open the *Pages* drop-down menu in the header and click on *Open Remote Controls Editor*
- In the *Device Pages* section click on the big *+* icon to create a new Remote Controls page





- Map the first 8 parameters of Ctrl-E to the 8 Remote Controls by clicking on the Wi-Fi icon in the upper left, then clicking on the first 8 parameters in the list to the right





- Save the mapping by clicking on *Save and Apply* in the Remote Controls editor
- The 8 macro parameters in Ctrl-E can now be controlled by the Osmose via the *adjust page*

Osmose as an External Instrument

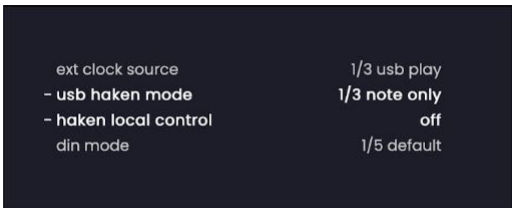
You can integrate Osmose's sound engine into Bitwig Studio to be able to record and play back performances as MIDI notes. Osmose's MIDI Port 2 must be used for this task exclusively.

1 Choose the Correct MIDI I/O Settings on Osmose

Head to Osmose's *midi i/o* tab. Set *usb haken mode* to *1/3 note only* to ensure Osmose sends MPE+ messages on Port 2.

Set *haken local control* to *off* to cut the direct connection between Osmose's keybed and its sound engine.

Important: With *haken local control* set to *off*, Osmose will only sound when its track in Bitwig Studio is armed.



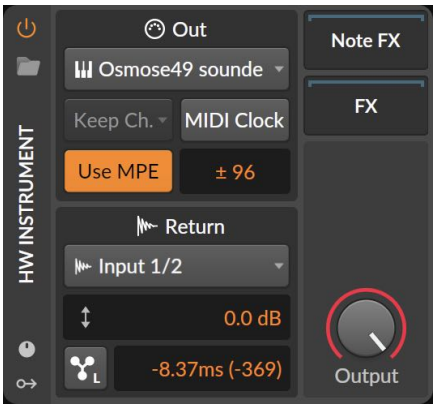
2 Integrating Osmose as an External Instrument in Bitwig Studio

Select an empty instrument track or create a new one. Add a HW Instrument device to the track.

Click on the input selector of the track in the inspector panel or the mixer and select MIDI Port 2 (*Osmose 49/61 sound engine* on macOS or *MIDIIN2 (Osmose 49/61)* on Windows) as the input. If the default setting *All ins* is not changed, controller data will be recorded twice, because Osmose sends data on both of its ports simultaneously.

On the HW Instrument device select Output 2 (*Osmose 49/61 sound engine* on macOS or *MIDIOUT2 (Osmose 49/61)* on Windows) in the MIDI output menu, and *Keep Ch.* in the MIDI Channel menu. Activate *Use MPE* and set the MPE Pitch Bend Range to *+/- 96* semitones.

Select the audio inputs your Osmose is connected to in the Return menu to receive Osmose's audio output on the same track. Arm the track, and you are ready to record, edit, and play back Osmose performances in Bitwig Studio.



Please note that overdubbing notes to an existing MIDI clip meant for Osmose's internal sound engine does not work reliably. The EaganMatrix in Osmose accepts MIDI notes only on MPE Note Channels 2 to 12, whereas the HW Instrument device is set to 16 by default.

As a workaround you can use the *Keep Ch.* option in the HW Instrument device: Applying no channel re-assignment at all omits note dropouts when replaying your single MPE clip, but it comes at the expense of losing overdub functionality.